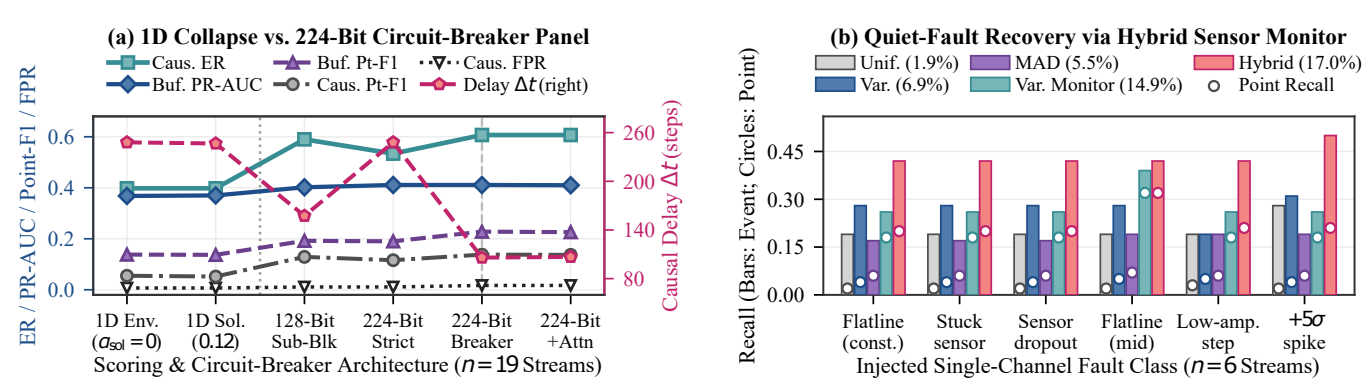

(a) 1D Collapse vs. 224-Bit Circuit-Breaker Panel
Caus. ER
Buf. PR-AUC
Buf. Pt-F1
Caus. Pt-F1
Caus. FPR
Delay Δt (right)
ER / PR-AUC / Point-F1 / FPR
Causal Delay Δt (steps)
0.0
0.2
0.4
0.6
80
140
200
260
1D Env. (σsol = 0)
1D Sol. (0.12)
128-Bit Sub-Blk
224-Bit Strict
224-Bit Breaker
224-Bit +Attn
Scoring & Circuit-Breaker Architecture (n = 19 Streams)
(b) Quiet-Fault Recovery via Hybrid Sensor Monitor
Unif. (1.9%)
MAD (5.5%)
Hybrid (17.0%)
Var. (6.9%)
Var. Monitor (14.9%)
Point Recall
Recall (Bars: Event; Circles: Point)
0.00
0.15
0.30
0.45
Flatline (const.)
Stuck sensor
Sensor dropout
Flatline (mid)
Low-amp. step
+5σ spike
Injected Single-Channel Fault Class (n = 6 Streams)